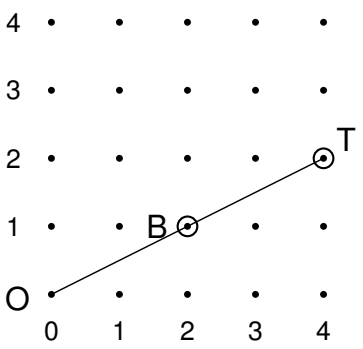
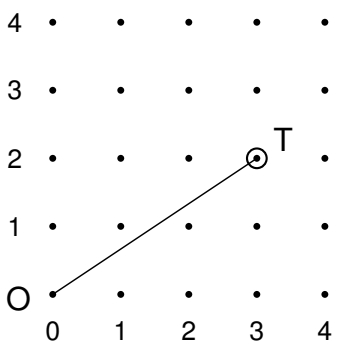
Stretch a thread from O to another dot T. A dot exactly on the thread between O and T blocks T. B marks a blocker. Keep O fixed while you test.

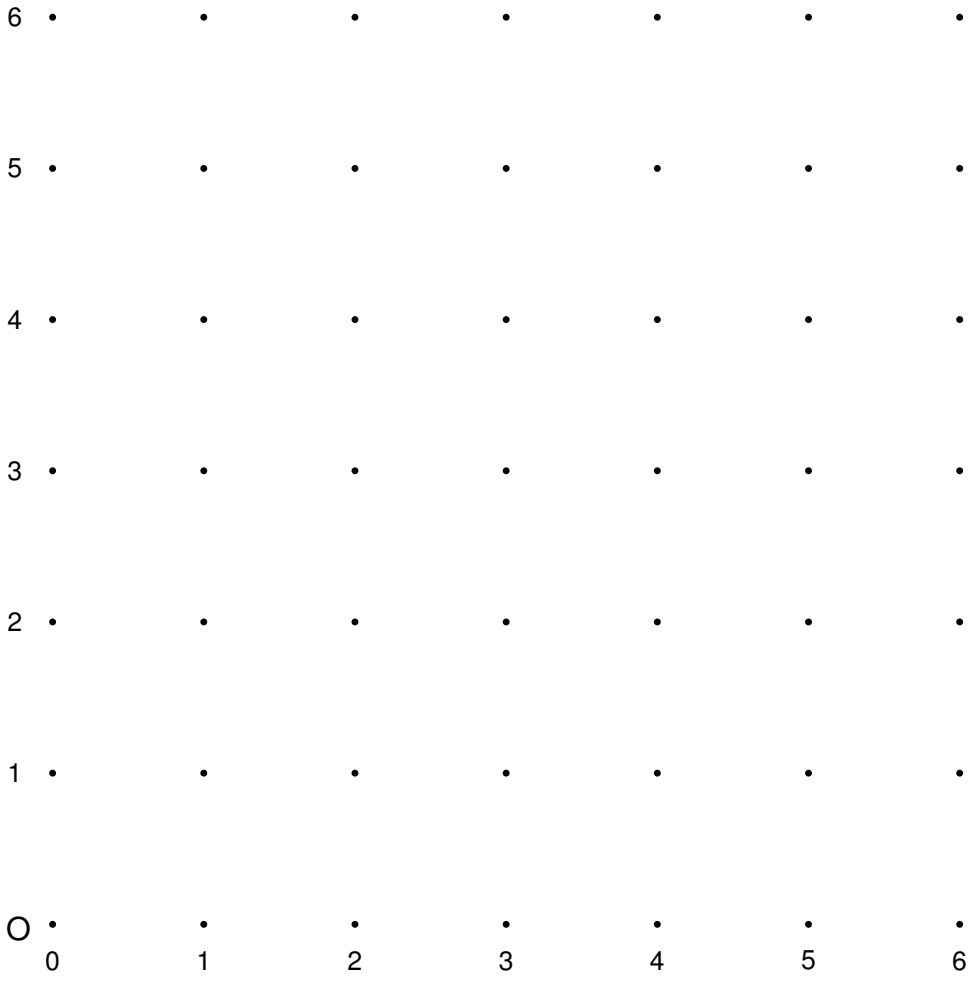


hidden

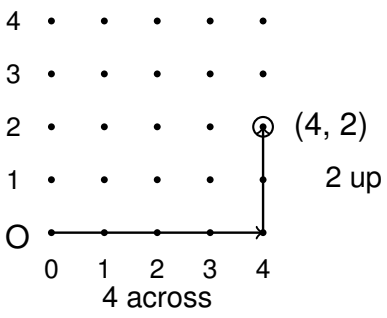


visible

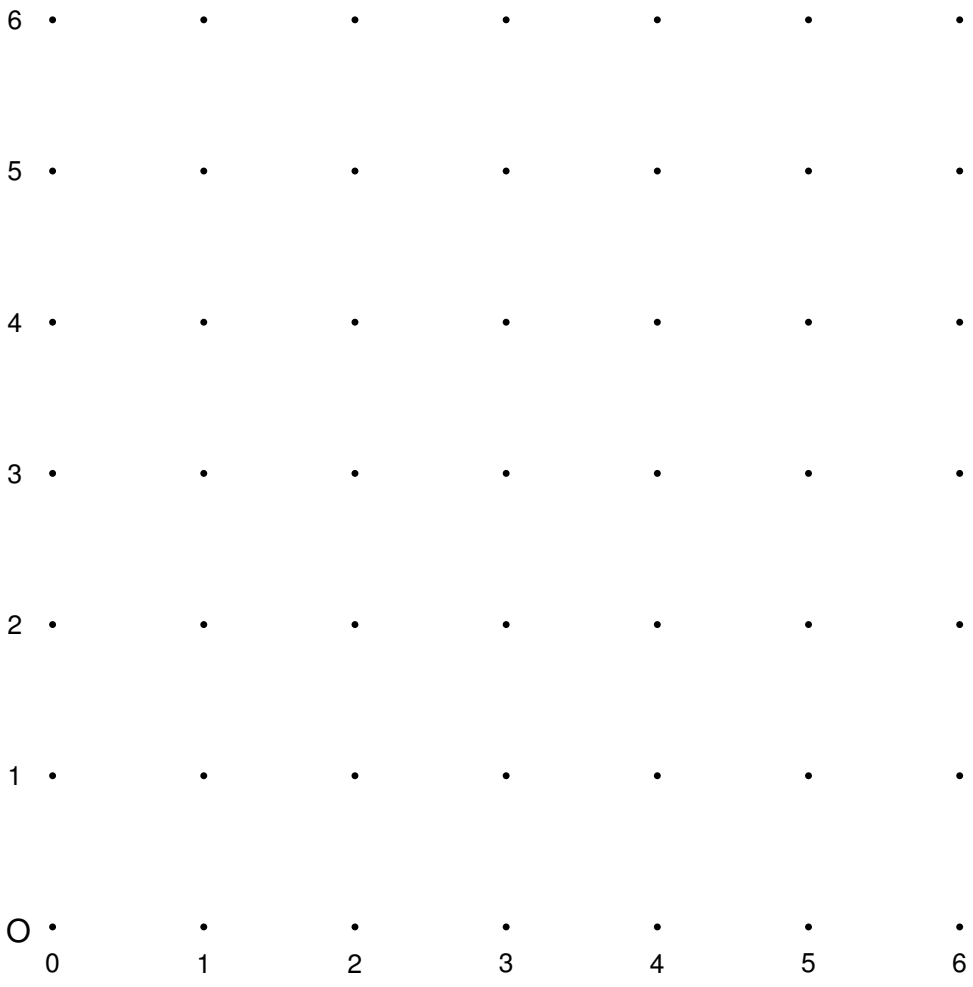
Problem 1: Circle every dot visible from O. Then find hidden targets with as many different numbers of blockers as you can, and mark each blocker.



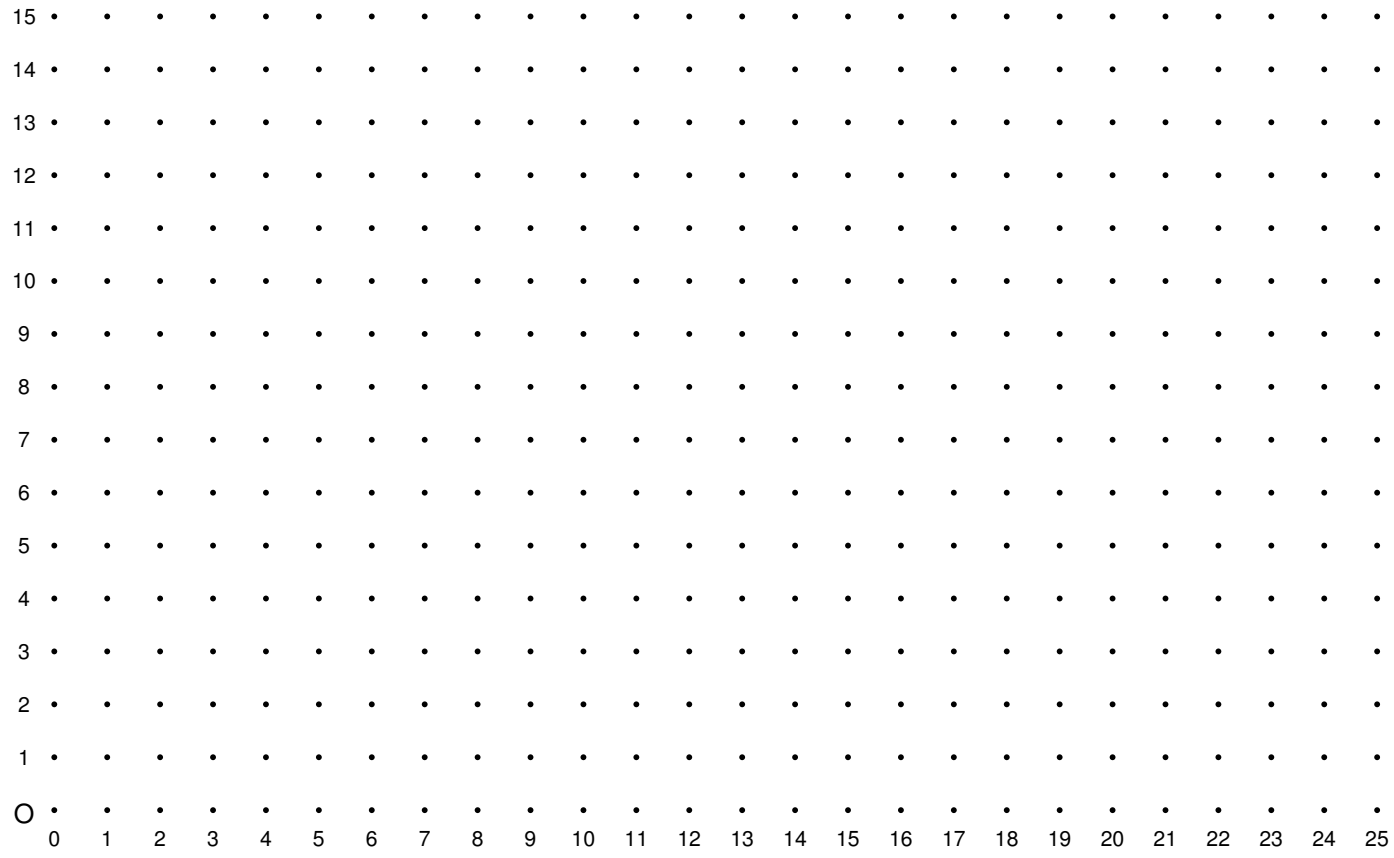
(4, 2) means 4 across and 2 up from O.



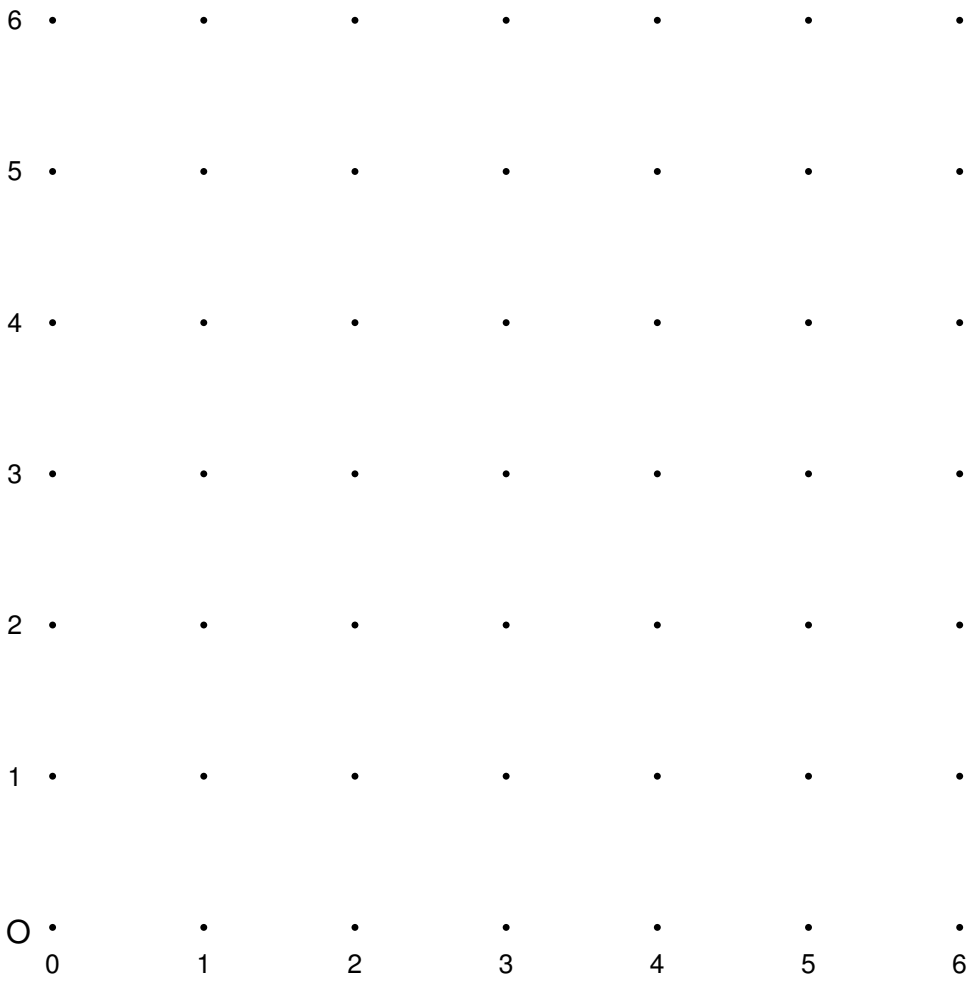
Problem 2: For each target, find the first dot after O on its line and every blocker: (6, 4), (6, 3), (5, 3), (6, 6), (6, 0) and (0, 6).



Problem 3: Predict the first dot and number of blockers for (12, 8), (15, 10), (21, 14), (25, 15) and (13, 8). Explain a rule that works for any target with whole-number coordinates.



Problem 4: List all the targets (a, b) whose two coordinates are each from 1 through 6 that have the same first visible dot as (6, 4). Repeat for (6, 3) and (6, 6). Explain why no target can belong to two different groups.



Problem 5: A target is (a, b), with a and b positive whole numbers. Explain why a common factor larger than 1 always produces a blocker. Does every blocker produce a common factor?

Problem 6: Explain why your Problem 3 rule finds every blocker: why can no other dot lie on the thread between O and the target?

Problem 7: Find an entire row of visible dots, an infinite collection of hidden dots, and a target with exactly 99 blockers. Explain each choice.